

CEH v13 MOCK EXAM 2

Knowledge Exam · 312-50

125 questions · 4 hours · single best answer

Target benchmark: 106/125 (≈85%)

Instructions

- **125 questions · 4 hours.** Pace $\approx$ 1 min 55 sec per question.
- Each question has exactly **one best answer**. Choose the single best option.
- No negative marking — never leave a question blank; flag and return if unsure.
- Answers are in the separate Answer Keys PDF. Do not open it until you finish.

Questions

- Q1.** Which security principle ensures that information is not altered by unauthorized parties?
- A. Integrity
 - B. Non-repudiation
 - C. Confidentiality
 - D. Availability
- Q2.** A hacker who breaks into systems to promote a political or social cause is best described as a:
- A. White hat
 - B. Script kiddie
 - C. Hacktivist
 - D. Suicide hacker
- Q3.** An auditor is given valid user credentials and partial network diagrams to simulate a malicious insider. Which test type is this?
- A. Gray-box
 - B. White-box
 - C. Blind
 - D. Black-box
- Q4.** Management asks which document type specifies mandatory, enforceable rules (e.g., 'passwords must be 14+ characters'). Which should you cite?
- A. Procedure
 - B. Standard
 - C. Policy
 - D. Guideline
- Q5.** In the STRIDE threat model, the 'E' stands for which threat?
- A. Encryption failure
 - B. Exfiltration
 - C. Elevation of privilege
 - D. Eavesdropping
- Q6.** Which tool is used to look up domain registration details such as registrar and name servers?
- A. WHOIS
 - B. Ettercap
 - C. Nmap
 - D. Hydra
- Q7.** A tester enters interactive mode and runs 'set type=MX' before querying a domain. Which tool is being used?
- A. dig in batch mode
 - B. nslookup
 - C. netstat
 - D. traceroute

Q8. Which Google dork operator restricts results to a single website or domain?

- A. filetype:
- B. site:
- C. inurl:
- D. intitle:

Q9. A traceroute stops returning hops partway to the target, and the last reply is an ICMP Type 3, Code 13. What does this indicate?

- A. The TTL was set too high
- B. The host is offline
- C. A routing loop
- D. Traffic is administratively prohibited by a firewall/ACL

Q10. Which email-security technology lets a domain owner specify which servers are authorized to send mail for the domain?

- A. SPF
- B. PGP
- C. DKIM
- D. DMARC

Q11. A reconnaissance report lists exposed internet-connected ICS devices found without scanning the target directly. Which service was most likely used?

- A. Hydra
- B. Shodan
- C. Wireshark
- D. Metasploit

Q12. Which Nmap option enables service and version detection?

- A. -PR
- B. -sn
- C. -sV
- D. -O

Q13. A tester needs a reliable scan that completes the full TCP handshake and does not require root privileges. Which scan fits?

- A. SYN scan (-sS)
- B. XMAS scan (-sX)
- C. ACK scan (-sA)
- D. Connect scan (-sT)

Q14. Which set of flags is sent in an Nmap XMAS scan?

- A. SYN only
- B. SYN, ACK
- C. FIN, PSH, URG
- D. ACK only

Q15. An ACK scan returns RST for a port. How should the tester interpret this?

- A. The port is closed and running no service
- B. The port is unfiltered
- C. The port is open
- D. The host is down

Q16. To reduce the chance of detection, a tester launches a scan with several decoy source addresses mixed with the real one. Which Nmap option is used?

- A. -f
- B. -sn
- C. -D
- D. -Pn

Q17. A captured packet from a network router shows a TTL near 255. Which device family is most consistent with this value?

- A. Linux server
- B. Android phone
- C. Windows desktop
- D. Networking equipment (e.g., Cisco)

Q18. Which scan type is best suited to discovering services like SNMP and DNS that do not use TCP?

- A. UDP scan (-sU)
- B. TCP SYN scan
- C. NULL scan
- D. ACK scan

Q19. Which protocol/port is commonly targeted for enumerating Windows shares and domain information?

- A. SMB / 445
- B. NTP / 123
- C. HTTP / 80
- D. SMTP / 25

Q20. An administrator runs 'nbtstat -A 10.0.0.15' against a host. What is being retrieved?

- A. The routing table
- B. The ARP cache
- C. Open TCP ports
- D. The remote NetBIOS name table by IP

Q21. In SNMP, what does the MIB represent?

- A. The database/hierarchy of managed objects
- B. The community string
- C. A single managed object
- D. The trap destination

Q22. A tester wants to enumerate Active Directory users and groups over LDAP from Linux. Which tool is MOST appropriate?

- A. hping3
- B. ldapsearch
- C. showmount
- D. macof

Q23. Which SMTP command expands a mailing list to reveal its members where supported?

- A. RCPT TO
- B. EXPN
- C. VRFY
- D. HELO

Q24. During enumeration, a tester finds SNMP responding to the community string 'public'. What access does this typically grant?

- A. Administrative shell
- B. Read-write
- C. No access
- D. Read-only

Q25. Which organization maintains the CVE list of vulnerability identifiers?

- A. NIST
- B. MITRE
- C. FIRST
- D. OWASP

Q26. Which risk-treatment option transfers the financial impact of a risk to a third party, such as through insurance?

- A. Avoidance
- B. Mitigation
- C. Transfer
- D. Acceptance

Q27. A vulnerability scanner misses a genuine, exploitable flaw entirely. Which outcome is this, and why is it serious?

- A. True positive; confirms the flaw
- B. False negative; leaves a real risk unaddressed
- C. True negative; expected result
- D. False positive; wastes analyst time

Q28. Which database, maintained by NIST, aggregates CVE entries along with CVSS scores, CWE, and CPE data?

- A. Exploit-DB
- B. GHDB
- C. NVD
- D. CAPEC

- Q29.** An assessment must map a finding to a standardized weakness category (e.g., improper input validation) rather than a specific product bug. Which standard applies?
- A. CVE
 - B. CWE
 - C. CVSS
 - D. CPE
- Q30.** Which tool is designed for GPU-accelerated offline password-hash cracking?
- A. Nikto
 - B. Hydra
 - C. Hashcat
 - D. Responder
- Q31.** Where are Active Directory domain account hashes stored on a domain controller?
- A. /etc/shadow
 - B. LSASS.dmp
 - C. Local SAM
 - D. NTDS.dit
- Q32.** An attacker forges a service ticket for a single service using that service account's hash. Which attack is this?
- A. Golden ticket
 - B. Pass-the-hash
 - C. Silver ticket
 - D. Kerberoasting
- Q33.** A pen tester distinguishes two password attacks: one tries one common password against many accounts to avoid lockouts. Which technique is described?
- A. Password spraying
 - B. Brute force
 - C. Credential stuffing
 - D. Rainbow table
- Q34.** Which classic weakness makes LM hashes easy to crack?
- A. They use SHA-512
 - B. They require a 20-character minimum
 - C. The password is uppercased and split into two 7-character halves with no salt
 - D. They are salted per user
- Q35.** After exploitation, an attacker clears Windows Security, System, and Application event logs. Which phase of system hacking is this?
- A. Privilege escalation
 - B. Covering tracks
 - C. Gaining access
 - D. Maintaining access

Q36. On Linux, an attacker abuses a binary that has the SUID bit set and is owned by root to run commands as root. What is being exploited?

- A. A cron job
- B. PATH injection only
- C. A SUID misconfiguration enabling privilege escalation
- D. A kernel race condition

Q37. Which malware disguises itself as legitimate software and does NOT self-replicate?

- A. Worm
- B. Rootkit
- C. Trojan
- D. Virus

Q38. Which malware component's sole job is to deliver and install other malware onto the target?

- A. Obfuscator
- B. Dropper
- C. Crypter
- D. Keylogger

Q39. An analyst inspects a malware sample's strings, PE headers, and imports WITHOUT executing it. Which technique is this?

- A. Sandboxing
- B. Static analysis
- C. Dynamic analysis
- D. Live forensics

Q40. Ransomware encrypts a victim's files, then the operators also threaten to publish the stolen data unless paid. Which model is this?

- A. Crypto-mining
- B. Double extortion
- C. Locker ransomware
- D. Scareware

Q41. Which virus type infers from its name that it changes its own code to evade signature detection, rewriting itself with each infection?

- A. Resident virus
- B. Macro virus
- C. Metamorphic virus
- D. Boot sector virus

Q42. A trusted software vendor's update mechanism is compromised to distribute malware to all customers. Which delivery method is this?

- A. Drive-by download
- B. Watering hole
- C. Malvertising
- D. Supply-chain attack

- Q43.** Which pairing correctly matches the malware technique to its goal?
- A. Injector: display advertisements
 - B. Packer: compress/obscure the executable to evade AV
 - C. Dropper: detect virtual machines
 - D. Downloader: encrypt files for ransom
- Q44.** Which type of sniffing only listens to traffic without injecting or altering packets?
- A. Active sniffing
 - B. MAC flooding
 - C. ARP spoofing
 - D. Passive sniffing
- Q45.** ARP is used to resolve which mapping on a local network?
- A. Domain name to IP
 - B. Port to service
 - C. MAC address to switch port
 - D. IP address to MAC address
- Q46.** On a fully switched network, why can't an attacker simply sniff all traffic by default?
- A. Switches encrypt all frames
 - B. A switch forwards unicast frames only to the destination port
 - C. Promiscuous mode is illegal
 - D. ARP is disabled on switches
- Q47.** An attacker wants a persistent man-in-the-middle position on a switched LAN to read a victim's web traffic. Which technique achieves this at Layer 2?
- A. ARP cache poisoning
 - B. Session fixation
 - C. SQL injection
 - D. DNS cache poisoning
- Q48.** Which Layer 2 control binds a limited number of MAC addresses to a switch port to resist MAC flooding?
- A. Dynamic ARP Inspection
 - B. DNSSEC
 - C. DHCP snooping
 - D. Port security
- Q49.** A capture filter versus a display filter in packet analysis differ how?
- A. They are identical
 - B. Capture filters limit what is recorded; display filters only hide/show captured packets
 - C. Capture filters hide packets after capture; display filters limit what is recorded
 - D. Display filters require root
- Q50.** A wireless NIC must capture all 802.11 frames in the air, including those for other devices. Which mode is required?
- A. Managed mode
 - B. Master mode
 - C. Monitor mode
 - D. Promiscuous mode

Q51. Searching through an organization's discarded trash for sensitive documents is which social-engineering technique?

- A. Baiting
- B. Shoulder surfing
- C. Tailgating
- D. Dumpster diving

Q52. Which attack leaves malware-laden USB drives in public areas hoping victims plug them in out of curiosity?

- A. Baiting
- B. Quid pro quo
- C. Pretexting
- D. Vishing

Q53. An attacker telephones a target pretending to be the bank's fraud department to extract account details. Which attack is this?

- A. Angler phishing
- B. Vishing
- C. Smishing
- D. Whaling

Q54. A phishing campaign specifically targets a company's CFO with a tailored message. Which term best fits?

- A. Mass phishing
- B. Smishing
- C. Pharming
- D. Spear phishing of an executive (whaling)

Q55. Which countermeasure most directly reduces the success of credential phishing even if a password is stolen?

- A. Disabling SPF
- B. Allowing password reuse
- C. Longer session timeouts
- D. Multi-factor authentication

Q56. Which CIA property is the primary target of a denial-of-service attack?

- A. Confidentiality
- B. Non-repudiation
- C. Availability
- D. Integrity

Q57. In a Smurf attack, which protocol is abused with a spoofed source and a broadcast address?

- A. TCP
- B. ARP
- C. ICMP
- D. UDP

Q58. A botnet operator issues commands to thousands of infected hosts through a central server to launch a coordinated flood. What is that server called?

- A. Honeypot
- B. Proxy chain
- C. Sinkhole
- D. Command-and-Control (C2)

Q59. Defenders redirect attack traffic to a controlled system to analyze it rather than simply dropping all of it. Which technique is this?

- A. Sinkholing
- B. Geo-blocking
- C. Blackholing
- D. Rate limiting

Q60. Which attack sends a packet with the same source and destination IP/port to confuse a target?

- A. LAND attack
- B. Ping of Death
- C. Fraggle
- D. Teardrop

Q61. An HTTP flood overwhelms a web application with seemingly legitimate requests. Why is it hard to mitigate?

- A. It only works over UDP
- B. It uses malformed packets easily dropped by firewalls
- C. It requires physical access
- D. The requests resemble normal user traffic

Q62. Which cookie attribute helps mitigate CSRF by restricting cross-site sending of the cookie?

- A. HttpOnly
- B. Secure
- C. Path
- D. SameSite

Q63. At which layer does stealing an HTTP session cookie to impersonate a user operate?

- A. Transport reset only
- B. Physical-level hijacking
- C. Application-level hijacking
- D. Network-level hijacking

Q64. A tester captures a user's session token over unencrypted HTTP and reuses it to access the account. Which protection would have prevented this?

- A. Enforcing HTTPS/TLS for all traffic
- B. Longer usernames
- C. Using HTTP instead of HTTPS
- D. Disabling the firewall

- Q65.** In a blind TCP session hijacking attempt, what is the attacker's main challenge?
- A. Cracking the TLS certificate
 - B. Reading the victim's screen
 - C. Correctly guessing the sequence numbers without seeing traffic
 - D. Bypassing the CAPTCHA
- Q66.** Which practice best protects active sessions against reuse if a token is somehow exposed?
- A. Storing tokens in the URL
 - B. Short session timeouts and token rotation
 - C. Using HTTP only
 - D. Never expiring sessions
- Q67.** Which device is typically deployed inline so it can actively drop malicious packets?
- A. Honeypot
 - B. SIEM
 - C. IDS
 - D. IPS
- Q68.** A detection method that matches traffic against a database of known attack patterns is called:
- A. Protocol tunneling
 - B. Anomaly-based
 - C. Heuristic baseline
 - D. Signature-based
- Q69.** An attacker encapsulates command traffic inside DNS queries to slip past a restrictive firewall. Which technique is this?
- A. ARP spoofing
 - B. Port knocking
 - C. DNS tunneling
 - D. Fragmentation
- Q70.** An IDS fails to alert on an actual intrusion. In the detection matrix, which outcome is this and why is it the worst for defenders?
- A. False positive; it wastes time
 - B. False negative; the real attack goes undetected
 - C. True positive; attack detected
 - D. True negative; nothing happened
- Q71.** Which firewall type makes decisions based on the state of active connections rather than inspecting each packet in isolation?
- A. Hub
 - B. Packet-filtering firewall
 - C. Stateful inspection firewall
 - D. Circuit-level gateway

- Q72.** A honeypot is most clearly distinguished from an IDS by the fact that it:
- A. Blocks traffic inline
 - B. Is intentionally designed to attract and study attackers
 - C. Only inspects encrypted traffic
 - D. Replaces the firewall
- Q73.** Which of the following is a common web server security misconfiguration?
- A. Disabling unused services
 - B. Hiding server banners
 - C. Enabling directory listing and leaving default credentials
 - D. Using parameterized queries
- Q74.** A three-tier web architecture separates which components?
- A. Presentation, application (logic), and data tiers
 - B. DNS, DHCP, and NTP
 - C. Router, switch, firewall
 - D. CPU, RAM, and disk
- Q75.** A web server defacement replaces the homepage with the attacker's message. Which security property is most directly violated?
- A. Integrity
 - B. Availability
 - C. Confidentiality
 - D. Non-repudiation
- Q76.** A tester finds that appending %2e%2e%2f sequences to a URL returns the contents of /etc/passwd. Which attack and root cause apply?
- A. Directory traversal; improper input validation of file paths
 - B. XSS; unencoded output
 - C. CSRF; missing token
 - D. SQL injection; string concatenation
- Q77.** Which component of a web server stores the configuration files and server binaries rather than published site content?
- A. Document root
 - B. Web proxy
 - C. Virtual host
 - D. Server root
- Q78.** Which attack injects client-side script that executes in other users' browsers?
- A. Directory traversal
 - B. Command injection
 - C. Cross-site scripting (XSS)
 - D. SQL injection

- Q79.** Which XSS variant has the payload reflected immediately from the request, often via a crafted link, without being stored?
- A. Reflected XSS
 - B. DOM-based XSS
 - C. Persistent XSS
 - D. Stored XSS
- Q80.** An application includes a user-controlled filename directly in a server-side 'include' that then executes attacker-hosted code from a remote URL. Which flaw is this?
- A. Remote File Inclusion (RFI)
 - B. Local File Inclusion (LFI)
 - C. Open redirect
 - D. IDOR
- Q81.** A reviewer must classify a flaw where the app trusts unsigned software updates and deserializes untrusted objects. Which 2021 OWASP category fits best?
- A. A08 Software and Data Integrity Failures
 - B. A05 Security Misconfiguration
 - C. A07 Identification and Authentication Failures
 - D. A02 Cryptographic Failures
- Q82.** Which defense most directly addresses CSRF?
- A. Parameterized queries
 - B. Rate limiting
 - C. Output encoding
 - D. Anti-CSRF tokens and SameSite cookies
- Q83.** A tester must choose the term for abusing a server to make requests to an internal admin panel not reachable from the internet. Which is correct?
- A. CSRF
 - B. Clickjacking
 - C. Session fixation
 - D. SSRF
- Q84.** Which web-service architectural style commonly uses HTTP verbs (GET/POST/PUT/DELETE) and JSON?
- A. WSDL
 - B. SOAP
 - C. REST
 - D. UDDI
- Q85.** SQL injection fundamentally occurs because an application fails to separate what from what?
- A. Encryption from hashing
 - B. Client from server
 - C. TCP from UDP
 - D. Data from executable query structure

Q86. Which SQL injection class exfiltrates data through a separate channel such as a DNS lookup?

- A. Error-based
- B. In-band
- C. Boolean blind
- D. Out-of-band

Q87. A tester appends 'AND SLEEP(5)' and the page consistently takes five seconds longer to load. Which SQLi type is confirmed?

- A. UNION-based
- B. Time-based blind
- C. Error-based
- D. Second-order

Q88. Input is sanitized on submission but a stored value later triggers injection when used by another query. Which SQLi type is this?

- A. First-order (classic)
- B. Second-order (stored) SQLi
- C. Union-based
- D. Tautology

Q89. Why is a WAF considered a supplemental rather than primary SQLi control?

- A. It can be evaded by obfuscation and does not fix the vulnerable code
- B. It encrypts the database
- C. It only works on UDP
- D. It replaces input validation

Q90. Which command-line tool would a tester most likely use to enumerate databases via '--dbs' against a vulnerable parameter?

- A. hydra
- B. aircrack-ng
- C. nmap
- D. sqlmap

Q91. A tester evades a naive keyword filter that blocks the word SELECT by using mixed case and inline comments like SeL/**/eCT. What weakness does this exploit?

- A. TLS misconfiguration
- B. A salted hash
- C. A signature/keyword filter that does not normalize input
- D. A strong parameterized query

Q92. Which wireless security protocol uses AES-CCMP for encryption?

- A. Open
- B. WEP
- C. WPA2
- D. WPA (TKIP)

Q93. What is the BSSID of a wireless network?

- A. The access point's MAC address
- B. The Wi-Fi password
- C. The channel number
- D. The human-readable network name

Q94. An attacker sets up a fake AP with the same SSID as a corporate network to lure clients and capture credentials. Which attack is this?

- A. Jamming
- B. Evil twin
- C. Rogue DHCP
- D. WPS brute force

Q95. A network still has WPS enabled. Which tool/technique could recover the WPA2 passphrase by attacking the WPS PIN?

- A. Nikto scanning the SSID
- B. sqlmap against the AP
- C. John cracking the BSSID
- D. Reaver brute-forcing the WPS PIN

Q96. Which Bluetooth attack sends unsolicited messages to nearby devices?

- A. Bluesnarfing
- B. Bluejacking
- C. Bluebugging
- D. Bluesmacking

Q97. Why is MAC address filtering considered a weak wireless control?

- A. MAC addresses can be easily sniffed and spoofed
- B. It requires RADIUS
- C. It disables WPA2
- D. It blocks all legitimate users

Q98. A company deploys WPA3-Personal specifically to stop attackers who capture the handshake from cracking the passphrase offline. Which mechanism provides this protection?

- A. WEP shared key
- B. TKIP
- C. Simultaneous Authentication of Equals (SAE)
- D. WPS

Q99. Which practice is used to map and record Wi-Fi networks while driving around an area?

- A. Bluesnarfing
- B. War chalking
- C. War driving
- D. Evil twinning

Q100. Gaining privileged (superuser) control over an Android device by removing manufacturer restrictions is called:

- A. Jailbreaking
- B. Rooting
- C. Tethering
- D. Sandboxing

Q101. Which enterprise capability allows remote wipe of a lost corporate phone?

- A. Mobile Device Management (MDM)
- B. Bluetooth pairing
- C. NFC
- D. APK sideloading

Q102. A user installs an app from an unofficial store that requests excessive permissions and exfiltrates contacts. Which risk category does this represent?

- A. Network jamming
- B. Hardware fault
- C. Malicious/third-party app with excessive permissions
- D. Secure sandbox

Q103. On iOS, which jailbreak type does NOT persist after a reboot and requires re-running the exploit via a computer each time?

- A. Untethered
- B. Semi-untethered
- C. Permanent
- D. Tethered

Q104. Which control separates corporate data from personal data on a BYOD device?

- A. Promiscuous mode
- B. Containerization
- C. Jailbreaking
- D. Rooting

Q105. In industrial control systems, what does SCADA stand for in function?

- A. A cloud service model
- B. A secure file transfer protocol
- C. A wireless encryption standard
- D. Supervisory control and data acquisition for industrial processes

Q106. Which component in an ICS directly controls physical equipment such as actuators and motors?

- A. HMI
- B. PLC
- C. Historian
- D. Firewall

Q107. A famous worm sabotaged uranium-enrichment centrifuges by manipulating their PLCs. Which malware was this?

- A. Conficker
- B. Mirai
- C. WannaCry
- D. Stuxnet

Q108. An IoT vendor ships devices with the same admin password in firmware that users cannot change. Which OWASP IoT risk is this, and why is it severe?

- A. Lack of physical hardening; enables theft
- B. Insufficient privacy; leaks location
- C. Weak/hardcoded passwords; enables mass compromise like botnets
- D. Insecure update; blocks patches

Q109. Which IoT communication model has devices connect through a local intermediary before reaching the cloud?

- A. Device-to-gateway
- B. Device-to-device
- C. Device-to-cloud
- D. Back-end data sharing

Q110. In which cloud model does the provider deliver complete applications that the customer simply uses, such as webmail?

- A. IaaS
- B. PaaS
- C. SaaS
- D. On-premises

Q111. Which cloud actor acts as an intermediary that manages use, performance, and delivery of cloud services between consumer and provider?

- A. Cloud broker
- B. Cloud carrier
- C. Cloud auditor
- D. Cloud consumer

Q112. An attacker who compromises a tenant VM attempts to infer another tenant's secret keys by measuring shared CPU cache timing. Which attack class is this?

- A. Side-channel attack
- B. SQL injection
- C. Wrapping attack
- D. Man-in-the-cloud

Q113. An attacker steals a victim's cloud file-sync token and gains persistent access to their storage without ever knowing the password. Which attack is this?

- A. DNS poisoning
- B. Wrapping attack
- C. Cryptojacking
- D. Man-in-the-cloud (MITC)

- Q114.** Which technology shares the host OS kernel and is more lightweight than a full virtual machine?
- A. Container
 - B. Hypervisor
 - C. SAN
 - D. Bare-metal server
- Q115.** A security review of a Kubernetes cluster flags overly broad permissions letting a pod act as cluster admin. Which risk theme is this?
- A. Weak TLS ciphers
 - B. Overly permissive RBAC
 - C. Open S3 bucket
 - D. Hardcoded Wi-Fi PSK
- Q116.** Which cryptographic function is one-way and cannot be reversed to recover the input?
- A. A hash function (e.g., SHA-256)
 - B. Base64 encoding
 - C. RSA encryption
 - D. AES encryption
- Q117.** Which algorithm is used purely for secure key exchange rather than bulk encryption or signatures?
- A. AES
 - B. RC4
 - C. SHA-1
 - D. Diffie-Hellman
- Q118.** To verify a digitally signed document came from Alice and was not altered, which key does the verifier use?
- A. Alice's private key
 - B. The verifier's private key
 - C. A shared secret
 - D. Alice's public key
- Q119.** An organization wants to protect encrypted data captured today against future quantum decryption. Which threat are they addressing?
- A. Harvest-now, decrypt-later
 - B. Rainbow tables
 - C. Padding oracle
 - D. Birthday attack
- Q120.** Which PKI component is responsible for verifying a requester's identity before a certificate is issued?
- A. Registration Authority (RA)
 - B. Certification Authority (CA)
 - C. Validation Authority (VA)
 - D. Certificate Revocation List (CRL)

Q121. A browser warns that a site's certificate is not trusted because it was signed by the site itself. What kind of certificate is this?

- A. Extended-validation certificate
- B. Self-signed certificate
- C. Wildcard certificate
- D. CA-signed certificate

Q122. A legacy service negotiates SSL 3.0 and is exploited to decrypt traffic by downgrading the connection. Which class of attack and fix apply?

- A. Rainbow table; salting
- B. Birthday attack; longer hashes
- C. Protocol downgrade (e.g., POODLE); disable SSL and enforce TLS 1.2/1.3
- D. Side-channel; constant-time code

Q123. Which attack exploits the mathematics of hash collisions and underlies concerns about weak hash functions?

- A. Birthday attack
- B. Man-in-the-middle
- C. Replay attack
- D. Brute-force key search

Q124. In a blockchain, an attacker isolates a target node from honest peers so it sees only attacker-controlled data. Which attack is this?

- A. Eclipse attack
- B. Race attack
- C. Finney attack
- D. 51% attack

Q125. Which pair correctly matches cipher to type?

- A. AES = stream, RC4 = block
- B. AES = block, RC4 = stream
- C. RSA = symmetric, AES = asymmetric
- D. SHA-256 = cipher, AES = hash